

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Technical Presentation

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1504

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Criteria	Technical Content (1.5)	Structure and Flow (1)	Use of Visuals (0.75)	Communication Skills (1)	Professionalism (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Technical Presentation Topics

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

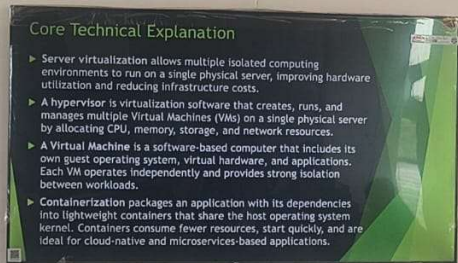
1. A cloud provider aims to improve data center efficiency. Present how hardware evolution (CPU virtualization support, GPUs, SSDs, high-speed networking) enhances cloud performance and scalability.
2. A company is redesigning its application using microservices and APIs. Present how the evolution from traditional web applications to cloud-native software has influenced modern application design.
3. A data center needs to maximize resource utilization. Prepare a technical presentation on server virtualization, covering hypervisors, virtual machines, and containerization. Compare their roles in cloud environments.
4. An enterprise integrates multiple distributed systems. Present how web services enable interoperability in cloud platforms, including standards, protocols, and service-oriented architecture (SOA).
5. An educational institution plans to adopt cloud services for labs, development, and administration. Present a comparative analysis of IaaS, PaaS, and SaaS, focusing on control, flexibility, cost, and use cases.

Minimum slide flow:

- Title and objective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway

Technical Presentation Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism



Core Technical Explanation

- ▶ Server virtualization allows multiple isolated computing environments to run on a single physical server, improving hardware utilization and reducing infrastructure costs.
- ▶ A hypervisor is virtualization software that creates, runs, and manages multiple Virtual Machines (VMs) on a single physical server by allocating CPU, memory, storage, and network resources.
- ▶ A Virtual Machine is a software-based computer that includes its own guest operating system, virtual hardware, and applications. Each VM operates independently and provides strong isolation between workloads.
- ▶ Containerization packages an application with its dependencies into lightweight containers that share the host operating system kernel. Containers consume fewer resources, start quickly, and are ideal for cloud-native and microservices-based applications.

GPS Map Camera

Kuthambakkam, Tamil Nadu, India



22g8+cwh, Kuthambakkam, Tamil Nadu 602105, India

Lat 13.026405° Long 80.017085°

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